

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE
A LEVEL 01 MODULE, AUTUMN SEMESTER 2020-2021
DATABASES AND INTERFACES
COMP1044

Time allowed: **TWO Hours**

Answer ALL questions

ADDITIONAL MATERIAL: Nil.

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules applies here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID**. Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

Question 1**[20 marks]**

- (a) The tables below are used by a company to record data about its employees and departments.

Table name: EMPLOYEE							
<u>EMPNO</u>	<u>ENAME</u>	<u>JOB</u>	<u>MGR</u>	<u>HIREDATE</u>	<u>SAL</u>	<u>COMM</u>	<u>DEPTNO</u>
7369	SMITH	CLERK	7902	17-DEC-90	800.00		20
7499	ALLEN	SALESMAN	7698	20-FEB-91	1600.00	300.00	30
7521	WARD	SALESMAN	7698	22-FEB-91	1250.00	500.00	30
7566	JONES	MANAGER	7839	02-APR-91	2975.00		20
7654	MARTIN	SALESMAN	7698	28-SEP-91	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	01-MAY-91	2850.00		30
7782	CLARK	MANAGER	7839	09-JUN-91	2450.00		10
7788	SCOTT	ANALYST	7566	09-NOV-91	3000.00		20
7839	KING	PRESIDENT		17-NOV-91	5000.00		10
7844	TURNER	SALESMAN	7698	08-SEP-91	1500.00	0.00	30
7876	ADAMS	CLERK	7788	23-SEP-91	1100.00		20
7900	JAMES	CLERK	7698	03-DEC-91	950.00		30

Table name: DEPARTMENT		
<u>DEPTNO</u>	<u>DNAME</u>	<u>LOC</u>
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Formulate SQL statements for the following queries:

- (i) Display the highest salary earned by an analyst.

[3 marks]

Suggested Answer:

```
SELECT MAX(sal)
FROM employee
WHERE job = 'analyst'
```

Marking Scheme:

Award 3 marks for correct answer, as stated above or any other acceptable answer.

Award a maximum of 1 mark for partial answer.

- (ii) Increase the salary of all clerks by 10%.

[2 marks]

Suggested Answer:

```
UPDATE employee
SET sal = 1.1 * sal
WHERE job = 'clerk'
```

Marking Scheme:

Award 2 marks for correct answer, as stated above or any other acceptable answer.

Award a maximum of 1 mark for partial answer.

(iii) List the names of all managers and the departments (**ename** and **dname**) they manage.

[2 marks]

Suggested Answer:

```
SELECT ename, dname
FROM employee e, department d
WHERE job = 'manager' AND e.deptno = d.deptno
```

OR

```
SELECT ename, dname
FROM employee e, department d
WHERE d.deptno IN
      (SELECT deptno FROM employee WHERE job = 'manager')
```

Marking Scheme:

Award 2 marks for correct answer, as stated above or any other acceptable answer.

Award a maximum of 1 mark for partial answer.

(iv) List the department name and the total number of employees in each department.

[4 marks]

Suggested Answer:

```
SELECT dname, COUNT(e.deptno) AS 'Total'
FROM employee e, department d
WHERE e.deptno = d.deptno
GROUP BY dname
```

Marking Scheme:

Award 4 marks for correct answer, as stated above or any other acceptable answer.

Award a maximum of 2 marks for partial answer.

(v) List the **ename** of clerks who are paid a higher salary than the clerk named 'James'. Use the 'sub query' concept.

[4 marks]

Suggested Answer:

```
SELECT ename
FROM employee
WHERE sal > (SELECT sal FROM employee
             WHERE job = 'clerk' AND ename = 'James') AND job = 'clerk'
```

Marking Scheme:

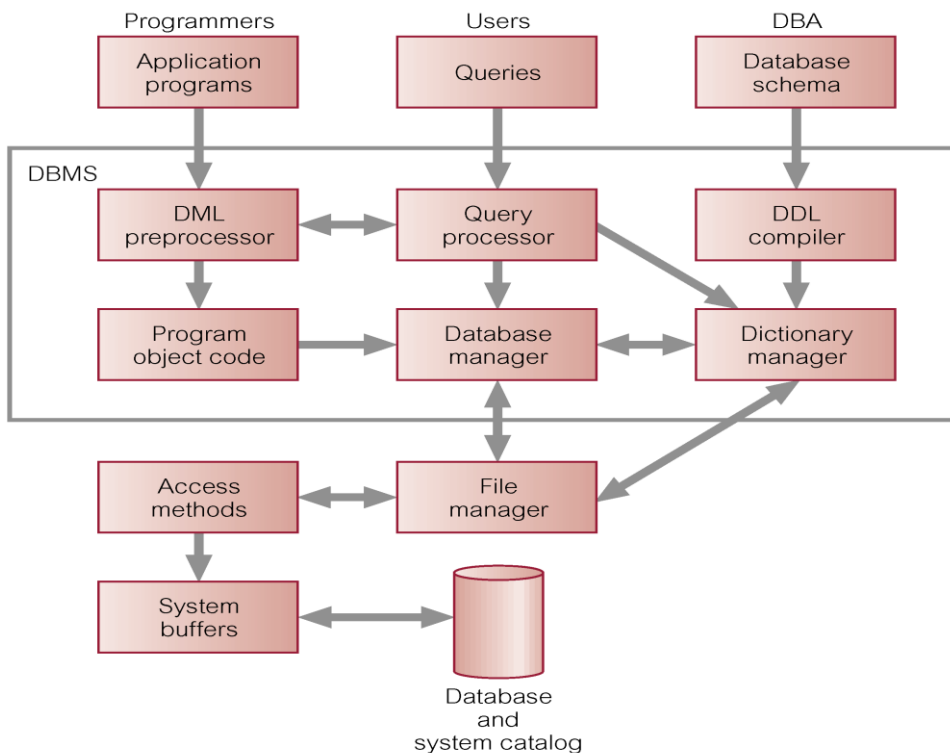
Award 4 marks for correct answer, as stated above or any other acceptable answer which used a sub-query.

Award a maximum of 2 marks for partial answer or any correct answer that does not use a sub-query.

(b)

Describe the **FIVE (5)** functional components of a Database Management System as shown in the diagram below:

[5 marks]



Suggested Answer:

The components:

- **Database Manager** – interfaces with user submitted application programs and queries.
- **Query Processor** – transforms queries into a series of low-level instructions directed to the Database Manager.
- **DML Preprocessor** – converts DML statements embedded in an application program into standard function calls in the host language.
- **DDL Compiler** – converts DDL statements into a set of tables containing meta-data.
- **Dictionary Manager** – manages access to and maintains the system catalog.

Marking Scheme:

Award 1 mark for the explanation provided for each component.

Maximum: 5 marks.

Question 2

[20 marks]

- (a) The IT Training Group (Kuala Lumpur) has contacted you to create a conceptual model by using the Entity-Relationship data model for a database that will meet the information needs for its training programme. The Company Director gives you the following description of the training group's operating environment.

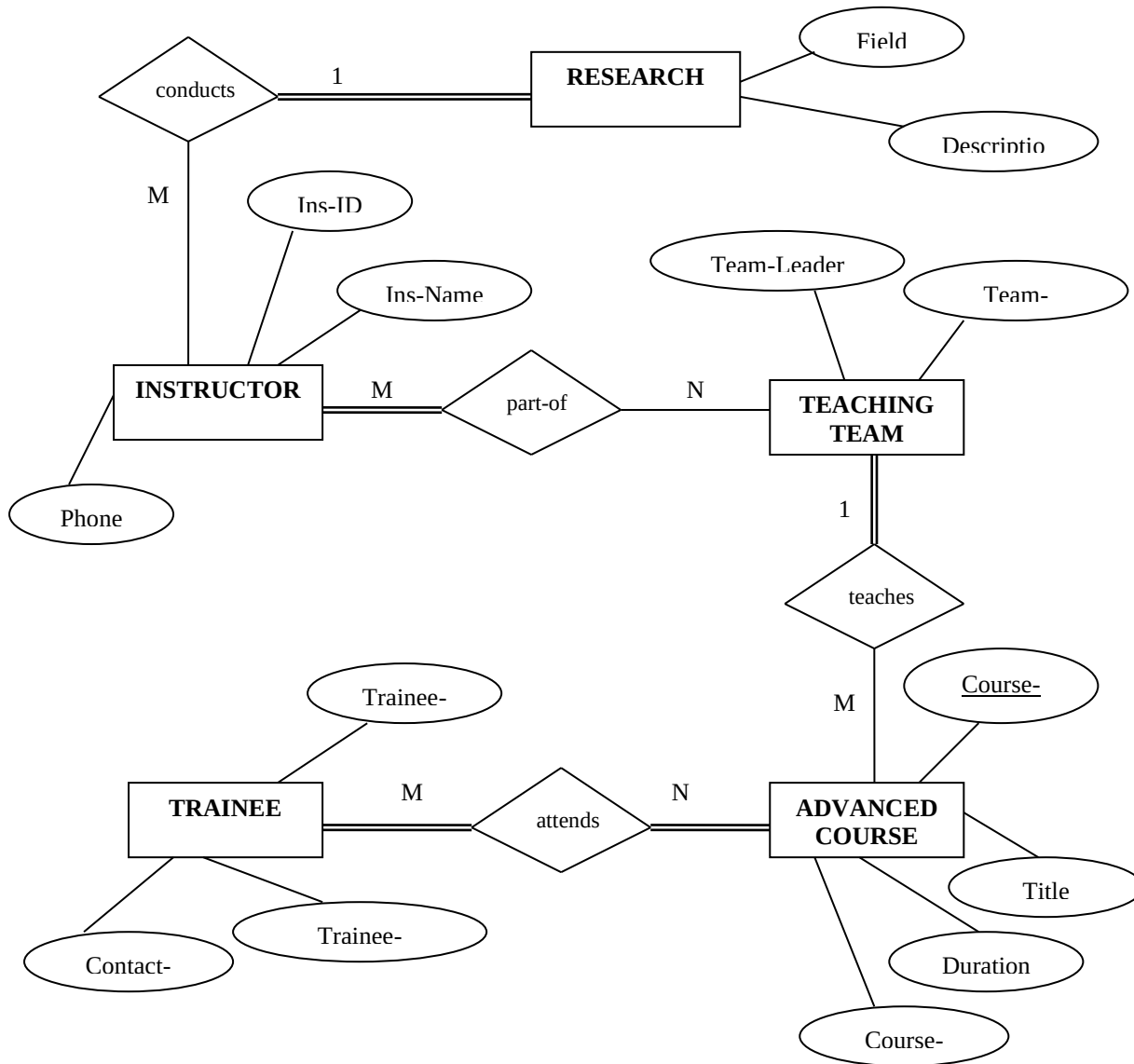
The Company has twelve instructors and can handle up to one hundred trainees per training session. The Company offers five advanced technology courses, each of which is taught by a teaching team of two or more instructors. Each instructor is assigned to a maximum of two teaching teams or may be assigned to do research in a particular

field. Each trainee undertakes one advanced technology course per training session.

From the enterprise rules above, draw an Entity Relationship (ER) diagram, with appropriate attributes, keys, cardinalities and participation constraints.

(10 marks)

Suggested Answer:



Marking Scheme:

Award:

3 marks for suitable entities

3 marks for suitable attributes and primary keys identified

3 marks for suitable relationships and participation constraints

1 mark if the overall diagram is correct.

Total: 10 marks.

b)

Suggested Answer:

- **Compactness/Centralised:** There is no need for possibly voluminous files. File based : Separation and isolation of data
- **Speed:** The DBMS can retrieve and update data faster. Embedded within one file
- **Concurrency Control:** Accurate, up-to-date information is available on demand at any time.
- **Protection:** The data can be better protected against unintentional loss and unlawful access in term of security system, integrity system, recovery control,
- Provides user-accessible catalog.
- **View mechanism:** Provides users with only the data they want or need to use.
- Enforcement of standards
- Data Sharing

Marking Scheme:

Award **2 marks** for any (5) benefits (as listed above or any other acceptable answer) explained, and contrast with file based approach.

Maximum: 10 marks.

Question 3

[20 marks]

- (a) Identify with example THREE (3) different types of errors in PHP.

[6 marks]

(6 marks)

Suggested Answer:

There are 3 types of error in PHP.

Notices: These are non-critical errors. These errors are not displayed to the users.

Warnings: These are more serious errors, but they do not result in script termination. By default, these errors are displayed to the user.

Fatal Errors: These are the most critical errors. These errors may cause due to immediate termination of script.

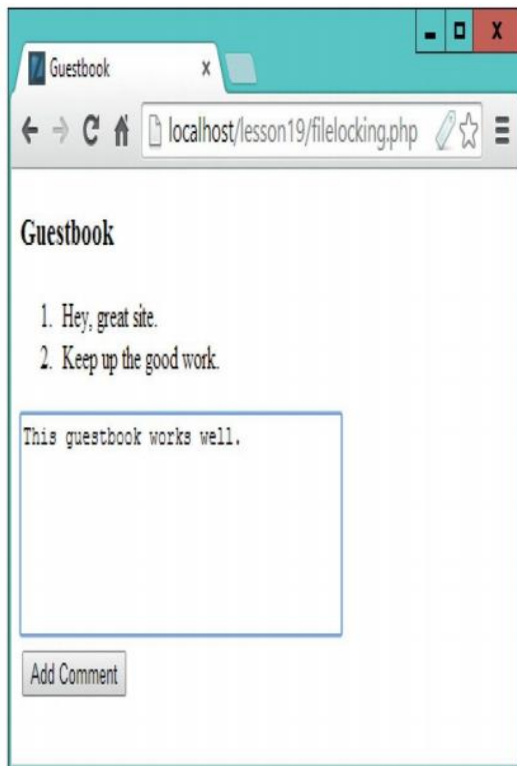
Marking Scheme:

Award:

2 marks for the name of each error in PHP system.

Maximum: 6 marks.

- (b) Develop the following output using php code as provided in figure below:



[6 marks]

Suggested Answer:

```
<?php
$message = '';
$self = $_SERVER['PHP_SELF'];
$guestbook = '';

if (isset($_POST['comment']))
{
    $comment = '<li>' . htmlentities($_POST['comment']) . "</li>\n";
    $handle = @fopen('guestbook.txt', 'a+')
        or $message = 'Cannot open file';

    if ($handle)
    {
        if (@flock($handle, LOCK_EX))
        {
            $flag = @fwrite($handle, $comment);
            @flock($handle, LOCK_UN);
            if ($flag == FALSE) $message = 'Cannot write to file';
        }

        @fclose($handle);
    }
}
```

```

    if ($flag == FALSE) $message = 'Cannot read file';
}

@fclose($handle);
}

echo <<<_EOT
<!DOCTYPE html>
<html>
  <head>
    <title>Guestbook</title>
    <style>
      textarea {
        width :300px;
        height:100px;
        resize:none;
      }
    </style>
  </head>
  <body>
    <h3>Guestbook</h3>
    <ol>$guestbook</ol>
    <form method='post' action='$self'>
      <textarea name='comment'></textarea><br>
      <input type='submit' value='Add Comment'>
    </form><br>
    $message
  </body>
</html>
_EOT;
?_

```

```

$handle = @fopen("guestbook.txt", 'r') != FALSE
or $message = 'Cannot open file';

```

```

if ($handle)
{
  $flag = TRUE;

  if (@flock($handle, LOCK_EX))
  {
    $length = @filesize('guestbook.txt');
    $guestbook = @fread($handle, $length) or $flag = FALSE;
    @flock($handle, LOCK_UN);
  }
}

```

Marking Scheme:

Award 2 marks for coding using correct syntax

Award 2 marks for file manipulations

Award 2 marks for communication with file update.

Total: 6 marks.

c) List out the different ways an HTML elements can be accessed in a JScript code.

- i. Gets an element by its ID name
- ii. Gets all the elements that have the given classname.
- iii. Gets all the elements that have the given tag name.
- iv. This function takes css style selector and returns the first selected element.

Suggested Answer:

(i) getElementById('idname'): Gets an element by its ID name

(ii) getElementsByClass('classname'): Gets all the elements that have the given classname.

(iii) getElementsByTagName('tagname'): Gets all the elements that have the given tag name.

(iv) querySelector(): This function takes css style selector and returns the first selected element.

Marking Scheme:

Award 2 marks for each correct JavaScript function – 4 x 2 = 8 marks

Question 4

[20 marks]

(a) Normalize the below Employee_Project table into Third Normal Form (3NF).

Project_No	Project_Name	Emp_No	Emp_Name	Hour	Chg_Hour	Job_Class
CR76	Payroll	PG4	Willy	35	85.00	Programmer
		PG16	Smith	45	78.00	DBA
CR56	Barcode	PG4	Willy	50	85.00	Programmer
		PG36	Kamal	37	65.00	Analyst
		PG16	Smith	45	78.00	DBA

[10 marks]

Suggested Answer:

INF :

Employee_No, Project_No -> Employee_Name, Project_Name, Hour, Chg_Hour, job_Class

OR

Employee_No -> Employee_Name, Job_Class, Chg_Hour

Project_No -> Project_Name

Job_Class -> Chg_Hour

2NF:

EMPLOYEE(Employee_No , Employee_Name, Job_Class, Chg_Hour)

PROJECT(Project_No , Project_Name)

ASSIGN (Project_No, Employee_No, Hour)

3NF:

EMPLOYEE(Employee_No , Employee_Name, Job_Class)
 PROJECT (Project_No , Project_Name)
 ASSIGN (Project_No, Employee_No, Hour)
 JOB(Job_Class, Chg_Hour)

Marking Scheme:

Award **3 marks** for each normal form (1NF, 2NF , 3NF) explained as stated above or any other acceptable answer. **Marking guideline as given below:**

Identification and removal of dependencies (partial & transitive) 1 marks

(5 x 2 = 10 marks)

Award maximum of **2 marks for any partial answer** on each normal form.

[Total:10 marks]

- (b) Establishing referential integrity using SQL: 2012 compliant database, explain the differences among the ON UPDATE RESTRICT, ON UPDATE CASCADE, and ON UPDATE SET NULL clauses, and results of declaring ON DELETE CASCADE.

[10 marks]

Marking scheme:

For each correct explanation 2.5 marks; max 10 marks

Model Answers

The SQL REFERENCES clause is used to establish referential integrity and prevents a foreign key value from being added if it is not already a valid value in the referenced primary key column.

The **ON UPDATE RESTRICT** clause allows a row to be deleted in a parent table only if no record in the child table references the primary key value to be deleted in the parent table.

The **ON UPDATE CASCADE** clause will cause a change in a primary key value of a parent table to be passed through and update the foreign key value in the related child table.

The **ON UPDATE SET NULL** option allows the update on the parent table, but changes the involved foreign key value in the child table to NULL.

Using the **SET NULL** option would result in losing the connection between the parent and child tables, not a desired effect.

Question 5

[20 marks]

- a) Employees and other insiders present the greatest database security problems to companies. Explain with suitable examples, any **FIVE (5)** potential threats posed by these personnel.

(10 marks)

Suggested Answer:

Any FIVE of the following:

- **Using another person's means of access**

- **Viewing and disclosing unauthorised data**
- **Blackmail**
- **Introduction of viruses**
- **Programmers creating trapdoors**
- **Program alterations**
- **Insufficiently trained staff**

Marking Scheme:

Award 1 mark for each threat (as listed above or any other acceptable answer) explained, plus 1 mark for each corresponding example. Maximum: 10 marks.

- b) An online bookstore will require additional security measures to protect its database transactions as opposed one which does not have an online presence. Describe any **TWO (2)** of such measures and explain using appropriate examples, how those measures can safeguard the bookstore's database.

[6 marks]

Suggested Answer:

Any TWO of the following:

- **Proxy servers**
- **Firewalls**
- **Message digest algorithms and digital signatures**
- **Digital certificates**
- **Kerberos**
- **Secure Socket Layer (SSL) and Secure HTTP (S-HTTP)**
- **Secure Electronic Transactions (SET) and Secure Transaction Technology (SST)**

Marking Scheme:

Award 1 mark for each measure (as listed above or any other acceptable answer) described plus 2 marks for each corresponding example on how that measure can be used in safeguarding the database. Maximum: 6 marks.

- (c) Briefly explain with example the following keys: [4 marks]

- candidate keys*
- primary key*

Suggested Answer:

- Candidate keys are selected keys in a relation used probably like search functions
- The primary key is the candidate key that is selected to identify tuples uniquely within a relation.

Marking Scheme:

Award **2 marks** for candidate keys explanation (as listed above or any other acceptable answer) explained.

Award **2 marks** for primary key explanation (as listed above or any other acceptable answer) explained.

Maximum: 4 marks.